

Memory as Inference: A Reconstruction-Centric Theory of Memory for Artificial Systems

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ABSTRACT

Artificial intelligence systems commonly model memory as storage and retrieval. This paradigm assumes that past states can be preserved and accessed with sufficient fidelity to support continuity. This assumption is structurally incomplete.

We argue that memory is not a stored artifact but a reconstructive inference process over partial, lossy, and context-dependent traces. Retrieval is not replay; it is constrained hypothesis formation. What systems present as “memory” is a context-conditioned reconstruction shaped by salience, prior confirmations, and uncertainty.

We formalize a three-tier representational model—trace, inference, and signal—and define constraints governing transitions between them. We introduce reconstruction as the core memory operation, derive invariants required for stability, and define memory drift as a measurable system property.

We show that treating inference as signal produces increasing drift in the absence of corrective mechanisms, and that modern AI systems already operate as unconstrained reconstruction systems. We further argue that current alignment approaches do not address memory-level instability.

We conclude that persistent AI systems must treat memory as inference or accept divergence in their internal representations over time.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; *Natural language processing*; • **Computer systems organization** → *Architectures*.

KEYWORDS

memory systems, reconstructive inference, AI alignment, memory drift, retrieval-augmented generation, long-context models

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1 INTRODUCTION

Persistent memory has become central to modern AI systems, including retrieval-augmented generation (RAG), long-context language models, and autonomous agents.

These systems implicitly rely on a shared assumption:

Memory is stored information that can be retrieved when needed.

However, observed system behavior contradicts this assumption:

- Outputs vary despite identical stored inputs
- Compressed representations change meaning across contexts
- Systems synthesize responses not explicitly encoded

These behaviors suggest a different model:

Memory is not retrieval—it is reconstruction under constraint.

This paper proposes a reconstruction-centric theory of memory for artificial systems and explores its implications for stability, interpretability, and alignment.

2 RELATED WORK

2.1 Cognitive Science (Contextual, Not Foundational)

Research in cognitive science has long characterized human memory as reconstructive [1, 8]. These findings provide useful analogies but are not required for the framework proposed here.

This theory does not depend on biological equivalence. Even if biological memory were perfectly reproductive, the behavior of modern AI systems still necessitates a reconstruction-based model.

2.2 Memory-Augmented Architectures

Graves et al. [3] introduced Differentiable Neural Computers, and Weston et al. [9] proposed Memory Networks. These systems extend memory capacity but retain retrieval-centric assumptions.

2.3 Retrieval-Augmented Systems

Lewis et al. [4] introduced retrieval-augmented generation (RAG), and Borgeaud et al. [2] extended this paradigm to trillion-token corpora. These approaches improve factual grounding but treat retrieved context as implicitly authoritative.

2.4 Recent Observations (2023–2025)

Recent work highlights instability in long-context and agent-based systems:

- Liu et al. [5]: degradation in long-context retrieval
- Mallen et al. [6]: hallucination persistence in RAG pipelines

- Park et al. [7]: generative agents exhibit memory drift over time

These findings align with the reconstruction-based model proposed here.

3 CONCEPTUAL MODEL OF MEMORY

We define memory as:

$$R_t = M(T_{\leq t}, C_t, S_{\leq t})$$

Where:

- T : traces (residual artifacts)
- C_t : current context at time t
- S : confirmed signals
- R_t : reconstructed representation at time t

These expressions are descriptive, not prescriptive. They define structural relationships rather than executable functions.

4 REPRESENTATIONAL ONTOLOGY

4.1 Trace (T)

Residual evidence of prior interaction: logs, embeddings, behavioral patterns.

Properties: incomplete; non-authoritative; semantically underdetermined.

Traces support meaning but do not define it.

4.2 Inference (I)

Hypothesis derived from traces.

Properties: probabilistic; context-dependent; revisable.

4.3 Signal (S)

Confirmed inference.

Properties: stable (within scope); persistent; authoritative.

4.4 Non-Collapse Principle

$$T \neq I \neq S$$

Each class must remain distinct. Collapse between classes is a primary failure mode.

5 THE CONFIRMATION PROBLEM

We define confirmation as:

$$S = C(I)$$

Where confirmation enforces: provenance, evidence, confidence, scope, and authority.

5.1 Inference Collapse

Definition. *Inference Collapse* occurs when probabilistic interpretations are treated as stable truth without confirmation.

This produces: drift, false personalization, and reinforced error.

5.2 Core Invariant

No inference may be treated as signal without confirmation.

6 RECONSTRUCTION

Memory is produced through reconstruction:

$$R = f(T, C, S)$$

Reconstruction involves: selection, weighting, structuring, and compression.

6.1 Reconstruction Invariants

- **Lineage** — outputs traceable to T or S
- **Non-Creation** — no unsupported meaning introduced
- **Uncertainty Preservation** — inference remains labeled as such
- **Signal Stability** — signals require explicit revision to change

7 SALIENCE

$$\Sigma = g(T, C, \Delta)$$

Salience is inferred from: recurrence, utility, identity relevance, affective weight, novelty, and contradiction.

Salience is not stored; it is recomputed at reconstruction time.

8 CONTRADICTION

Contradiction is a structural state within memory systems.

8.1 States

- **Stable** — consistent across traces
- **Disputed** — conflicting evidence present
- **Supersession Candidate** — prior state pending replacement

8.2 Preservation Principle

Contradiction should be preserved until resolution is justified.

Failure to preserve contradiction is itself a source of memory drift. When conflicting inferences are prematurely resolved or suppressed, systems lose information about uncertainty and alternative interpretations. This forces reconstruction to operate on artificially stabilized representations, increasing the likelihood of compounding error over time.

9 MEMORY DRIFT

9.1 Definition

Memory Drift is the divergence between reconstructed representations and their originating trace distribution over time.

9.2 Drift Theorem (Informal)

Systems that promote inference to signal without confirmation may exhibit increasing memory drift in the absence of corrective mechanisms such as external grounding, periodic revalidation, or explicit confirmation constraints.

Sketch:

- (1) Inference introduces error ϵ
- (2) Without confirmation, error becomes persistent
- (3) Future inference compounds existing error

This behavior is not purely theoretical. Park et al. [7] document divergence in memory representations within generative agents over extended simulations, where accumulated internal state deviates from initial conditions despite stable environmental inputs.

10 EMPIRICAL ALIGNMENT

Modern systems already exhibit reconstruction behavior:

RAG Systems. Mis-ranked retrieval can become implicitly authoritative [6].

Conversational Memory. Preferences are inferred and treated as stable across sessions.

Summarization. Compression introduces irreversible information loss [5].

Multi-Turn Dialogue. Context shifts reinterpret past interactions, altering effective meaning.

11 ALGORITHMIC IMPLICATIONS

The following pseudocode formalizes the reconstruction and confirmation operations described above.

Listing 1: Reconstruction

```
function reconstruct(T, C, S):
    relevant = select(T, C)
    weighted = apply_saliency(relevant)
    structured = organize(weighted)
    merged = integrate(structured, S)

    enforce_lineage(merged)
    enforce_uncertainty(merged)

    return merged
```

Listing 2: Confirmation

```
function confirm(I):
    require_provenance(I)
    require_evidence(I)
    require_confidence(I)

    assign_scope(I)

    return Signal(I)
```

12 LIMITATIONS AND FUTURE WORK

This paper presents a conceptual framework. Known limitations include:

- No empirical validation included
- Drift theorem is informal
- Saliency and confirmation functions are not fully specified

While reconstruction and confirmation are presented algorithmically, their mapping to current large-scale architectures remains non-trivial. Existing systems do not cleanly separate trace, inference, and signal representations, and often

lack explicit mechanisms for confirmation or contradiction preservation.

Future work includes: controlled drift experiments, quantitative saliency models, and system-level evaluation of memory stability.

13 IMPLICATIONS FOR ALIGNMENT

Current alignment approaches focus on outputs. This omits the evolution of internal representations over time.

13.1 Memory-Level Misalignment

Examples of memory-level failure include: incorrect user modeling, reinforced assumptions, and emotional misinterpretation. These are not output violations. They are memory failures.

As such, they are largely invisible to current alignment techniques, which operate on outputs rather than the evolving internal state that produces them.

14 DESIGN REQUIREMENTS

Stable persistent systems must:

- Separate T / I / S as distinct representational classes
- Enforce confirmation before promoting inference to signal
- Constrain reconstruction via the invariants in Section 6
- Preserve contradiction until resolution is justified
- Maintain lineage from outputs to originating traces or signals

15 IMPLICATIONS AND OPEN CHALLENGES

Memory governs long-term system coherence. Open challenges include:

- Scalable confirmation mechanisms
- Reliable saliency modeling
- Auditable memory evolution
- Translating conceptual models into deployable architectures

16 CONCLUSION

Memory is not storage. It is reconstruction, inference, and confirmation.

Systems that treat inference as signal drift. Systems that enforce the distinction between trace, inference, and confirmed signal remain coherent over time.

As persistent AI systems become more capable and more deeply integrated into high-stakes contexts, memory-level stability is not an implementation detail. It is a foundational property. This paper provides a conceptual framework toward that end.

REFERENCES

- [1] Frederic C. Bartlett. 1932. *Remembering: A Study in Experimental and Social Psychology*. Cambridge University Press.
- [2] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego De Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom

- Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, and Jack W. Rae. 2022. Improving Language Models by Retrieving from Trillions of Tokens. In *Proceedings of the 39th International Conference on Machine Learning*. 2206–2240.
- [3] Alex Graves, Greg Wayne, Malcolm Reynolds, Tim Harley, Ivo Danihelka, Agnieszka Grabska-Barwinska, Sergio Gómez Colmenarejo, Edward Grefenstette, Tiago Ramalho, John Agapiou, Adrià Puigdomènech Badia, Karl Moritz Hermann, Yori Zwols, Georg Ostrovski, Adam Cain, Helen King, Christopher Summerfield, Phil Blunsom, Koray Kavukcuoglu, and Demis Hassabis. 2016. Hybrid Computing Using a Neural Network with Dynamic External Memory. *Nature* 538, 7626 (2016), 471–476.
- [4] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, Vol. 33. 9459–9474.
- [5] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2023. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics* 12 (2023), 157–173.
- [6] Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*. 9802–9822.
- [7] Joon Sung Park, Joseph C. O’Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.
- [8] Daniel L. Schacter. 1999. The Seven Sins of Memory: Insights from Psychology and Cognitive Neuroscience. *American Psychologist* 54, 3 (1999), 182–203.
- [9] Jason Weston, Sumit Chopra, and Antoine Bordes. 2014. Memory Networks. *arXiv preprint arXiv:1410.3916* (2014).